KEERTHANA M

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LinkedIn | Github | Portfolio | Leetcode

SUMMARY

B.Tech Artificial Intelligence & Data Science graduate with hands-on experience in Machine Learning, Generative AI, NLP-related applications, and Python-based AI development. Experienced in building AI-powered applications using Python, FastAPI, scikit-learn, and Google Gemini API, including LLM-based conversational systems with contextual memory. Strong foundation in supervised and unsupervised learning, data preprocessing, model development, image processing, and REST API integration. Familiar with Generative AI, Large Language Models (LLMs), prompt engineering, and deep learning fundamentals, with a strong interest in developing and deploying practical AI solutions.

EDUCATION

B.Tech - Artificial Intelligence & Data Science - Agni College of Technology **2022 - 2026**
- CGPA: 8.6

HSC - St. Pius X Mat. Hr. Sec. School **2021 - 2022**
- Percentage: 90%

SSLC - St. Pius X Mat. Hr. Sec. School **2019 - 2020**
- Percentage: 90.8%

SKILLS

- **Programming:** Python, Java
- **Machine Learning & Deep Learning:** Supervised Learning, Unsupervised Learning, Model Evaluation, Feature Engineering, Data Preprocessing
- **Generative AI / NLP:** Generative AI, Large Language Models (LLMs), Google Gemini API, Prompt Engineering, NLP Fundamentals, Speech-to-Text, Contextual Conversation
- **Data & Scientific Computing:** NumPy, Pandas, Scikit-learn, Matplotlib, Tensorflow
- **Backend / APIs:** FastAPI, REST APIs, Spring Boot (Basics)
- **Frontend:** HTML, CSS, JavaScript, React.js
- **Database:** MySQL
- **Tools:** GitHub, Postman, VS Code
- **Core CS:** Data Structures & Algorithm, Operating System, OOPS, DBMS

EXPERIENCE

Research & Development Intern | Garuda Aerospace Ltd | Feb–Mar 2026
- Contributed to multispectral crop-imaging and agricultural monitoring workflows using vegetation-index analysis techniques such as NDVI.
- Assisted in developing backend data-processing pipelines and image-processing workflows for precision-agriculture applications.
- Worked with UAV-based agricultural data and analytical workflows for crop-health monitoring.

Machine Learning & Web Development Intern | Tamizhan Skills | Dec 2025
- Worked on machine learning concepts and backend-integrated web applications using Python and web technologies.
- Assisted in debugging, testing, and improving application functionality and performance.
- Contributed to implementing and integrating application features across backend and frontend components.

Data Science Intern | Lokey Future Technologies | Oct 2023–Apr 2024
- Performed data preprocessing, data cleaning, and exploratory data analysis using Python-based workflows.
- Automated repetitive data-processing tasks to improve workflow efficiency and reduce manual effort.

PROJECTS

Multispectral Paddy Crop Analysis System | Python, FastAPI, Machine Learning | Mar 2026

- Developed a full-stack AI application for UAV-based paddy crop monitoring using Python, FastAPI, React, and machine learning.
- Implemented data and image-processing workflows using NDVI, NDRE, and NDWI vegetation indices for crop-health analysis.
- Developed a Random Forest-based crop classification model and implemented model-driven analytical map generation.
- Built REST APIs and interactive dashboards for image processing, crop classification, analysis, and visualization.

Women Safety Web Application | Java, Spring Boot, MySQL | Sep 2025

- ·Developed a full-stack safety application using Spring Boot, Hibernate, MySQL, REST APIs, and JWT authentication.
- Implemented OTP verification, real-time location tracking using Geolocation API and Leaflet.js, and emergency SOS email alerts.
- Developed backend services for user management and emergency contact handling.

AI Assistant Chatbot using Gemini LLM and Speech Recognition | Python, FastAPI, Gemini API | Dec 2024

- Built an AI-powered conversational assistant using Google Gemini LLM, FastAPI, and browser speech APIs.
- Implemented speech-to-text, text-to-speech, and session-based contextual conversation memory for interactive user conversations.
- Developed REST APIs and integrated frontend and backend components for AI-powered conversational workflows.
- Deployed the application on Render and integrated the Gemini API for LLM-based response generation.

CERTIFICATIONS

- Introduction to Large Language Models (LLMs) | NPTEL | Jul–Oct 2025
- Introduction to Machine Learning | NPTEL | Jul–Oct 2025
- Deep Learning | NPTEL (Elite + Silver) | Jul–Oct 2024
- Data Science for Engineers | NPTEL (Elite + Gold) | Jul–Sep 2024
- Career Essentials in Software Development | Microsoft & LinkedIn | Mar 2025
- The Joy of Computing Using Python | NPTEL | Jan–Apr 2023

ACHIEVEMENTS

- Solved 400+ problems across Leetcode and Skillrack
- NPTEL Domain Scholar in Data Science – 2025
- NPTEL Discipline Star in Computer Science and Engineering – 2025
- 30th Topper – NPTEL Data Science for Engineers – 2024
- Awarded Outstanding Achiever – 2023